

Cloud Service Model

1. List three popular apps you use (like Instagram, Zomato, or Spotify) and for each, identify whether it is most similar to IaaS, PaaS, or SaaS. Briefly explain your reasoning for each choice.

ANS :

Application	Cloud Service Model	Reason
Instagram	SaaS (Software as a Service)	Instagram is a ready-to-use social media application. Users simply log in and use the app without managing servers, storage, or software updates.
Spotify	SaaS (Software as a Service)	Spotify is an online music streaming service. Users can stream songs and podcasts through the app without installing or maintaining any backend infrastructure.
Amazon EC2	IaaS (Infrastructure as a Service)	Amazon EC2 provides virtual servers in the cloud. Users are responsible for installing the operating system, applications, and managing their own server environment.

2. Create a table comparing IaaS, PaaS, and SaaS with at least three points for each model, using real-world examples such as AWS EC2, Google App Engine, and Office 365.

ANS :

Feature	IaaS (Infrastructure as a Service)	PaaS (Platform as a Service)	SaaS (Software as a Service)
Definition	Provides virtual infrastructure such as servers, storage, networking, and virtual machines.	Provides a complete platform for developing, testing, and deploying applications.	Provides fully functional software that users can access through the internet.
User Manages	Operating system, applications, middleware, runtime, and data.	Applications and data only.	Only uses the software; no technical management is required.

Feature	IaaS (Infrastructure as a Service)	PaaS (Platform as a Service)	SaaS (Software as a Service)
Provider Manages	Physical servers, storage, networking, and virtualization.	Hardware, operating system, middleware, runtime, networking, and security.	Everything including hardware, software, updates, maintenance, and security.
Advantages	High flexibility and full control over the infrastructure.	Faster application development without managing servers.	Easy to use, accessible from anywhere, and requires no installation or maintenance.
Real-World Example	AWS EC2	Google App Engine	Microsoft Office 365
Best Used For	Hosting websites, virtual machines, enterprise applications, and custom server environments.	Developing, testing, and deploying web or mobile applications.	Email services, office applications, online collaboration, and media streaming.

3. Sign up for a free AWS account (or use an existing one), navigate to the EC2 dashboard, and launch a new t2.micro EC2 instance running Ubuntu 22.04. Take a screenshot of your running instance in the AWS console.

Hint: Use the 'Launch Instance' button and select the free tier eligible Ubuntu AMI.

ANS :

- 1) Log in to the AWS Management Console.
- 2) Open the **EC2 Dashboard**.
- 3) Click **Launch Instance**.
- 4) Enter an instance name.
- 5) Select **Ubuntu Server 22.04 LTS** (Free Tier Eligible).
- 6) Choose **t2.micro** as the instance type.
- 7) Create or select an existing key pair.
- 8) Allow SSH (port 22) in the Security Group.
- 9) Click **Launch Instance**.
- 10) Wait until the instance state changes to **Running**.

4. After your EC2 instance is running, connect to it using SSH from your local terminal or an online SSH client. Run the command 'uname -a' and note the output.
Hint: You may need to use the .pem key file downloaded during instance setup.

ANS :

- 1) ssh -i mykey.pem ubuntu@<Public-IP>
 - Run:
 - uname -a
 - Example Output:
 - Linux ip-172-31-xx-xx 6.x.x-aws #xx-Ubuntu SMP x86_64 GNU/Linux
5. Suppose you want to host a simple static website for your college fest using your EC2 instance. List the basic steps you would follow after launching the instance to make your site accessible to everyone. Write 3-5 steps in order.

ANS :

- 1) Connect to the EC2 instance using SSH.
- 2) Install the Apache web server using:
 - sudo apt update
 - sudo apt install apache2 -y
- 3) Copy your website files
 - Ensure the EC2 Security Group allows inbound HTTP traffic on **port 80**.
 - Open the EC2 Public IP address in a web browser to verify that the website is accessible.